

Escaping loops

Programs can get stuck in a loop, but there are ways to escape. The word “break” leaves a loop (even a “forever” loop), and the word “continue” skips back to the start of the next loop.

SEE ALSO

◀ 122–123 Loops in Python

◀ 124–125 While loops

Inserting breaks

Putting a break into a loop makes the program jump out of the loop at once—even if the loop condition is true. Any commands inside the loop that come after the break are ignored.

1 Write a simple program

This program tests the user on the 7 multiplication table. The program continues looping until all 12 questions are answered. Write this program in the code window because it will be edited later.

```
table = 7
for i in range(1, 13):
    print('What\'s', i, 'x', table, '?')
    guess = input()
    ans = i * table
    if int(guess) == ans:
        print('Correct!')
    else:
        print('No, it\'s', ans)
print('Finished')
```

The variable “i” will count from 1 to 12

“i” is the loop variable

The backslash (“\”) tells Python the next quote mark is an apostrophe, not the end of the string

2 Insert a “break”

A “break” can be added so the user can escape the loop. The program executes a break if the user types “stop”.

If “guess” equals “stop”, the program skips the rest of the loop and prints “Finished”



```
table = 7
for i in range(1,13):
    print('What\'s', i, 'x', table, '?')
    guess = input()
    if guess == 'stop':
        break
    ans = i * table
    if int(guess) == ans:
        print('Correct!')
    else:
        print('No, it\'s', ans)
print('Finished')
```

The “ans” variable holds the correct answer to the question